From Raw Text to a Domain Knowledge Graph: Comparing Rule-Based and Off-the-Shelf NLP Extraction

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1. MOTIVATION

Knowledge Graphs (KGs) organize real-world entities and relationships in a structured way, enabling applications such as **search, recommendation, question answering**, and reasoning.

Automatic extraction from raw text is challenging due to the trade-off between precision and coverage.

Goal: Build a mini KG pipeline from movie plot summaries and compare two extraction strategies: **rule-based patterns** vs. **off-the-shelf NLP**, evaluating both coverage and precision.

2. DATASET

Wiki Movie Plots with Summaries
(Hugging Face dataset)
vishnuppriyavr/wiki-movie-plots-with-summaries

- **Source**: Wikipedia plot summaries
- **Language**: English
- **Number of plots** (original): ~45,000+

Preprocessing & Sampling

- Filtered by length: 250 – 2,500 characters
- Removed missing values
- Random sample of **250 plots** (seed = 42)
- Same sample used for both methods

3. EXPERIMENTAL SETUP

1. **Extraction** — Two strategies extract (**head, relation, tail**) triples from the same 250 plot summaries.
2. **Normalization** — Lowercasing, whitespace trimming, and basic canonicalization to reduce duplicates.
3. **Graph Construction** — Directed graphs built with NetworkX. Nodes = entities, edges = relations.
4. **Evaluation** — Coverage metrics computed automatically. Precision estimated via manual audit (10 triples per method).

4. METHODS

A. Rule-Based (Pattern Matching)

- **Entities**: capitalized words/phrases (simple NER heuristic)
- **Relations**: handcrafted patterns capturing common events.
  Examples: "X kills Y", "X meets Y", "X loves Y", "X travels to Y", "X finds Y", "X helps Y"
- **High** precision by design, limited recall

B. Off-the-Shelf NLP (textacy SVO)

- **Pipeline**: spaCy (en_core_web_sm) + textacy
- **Extraction**: Subject–Verb–Object triples from dependency parses
- Captures broader variety of relations
- **Higher** recall, potentially lower precision

Common Post-processing

- Remove duplicate triples (unique)
- Compute support count for each triple
- Build graphs and compute structural statistics

5. RESULTS – METRICS & EVALUATION

5.1 Coverage (Extraction Statistics)

Method	Raw Extractions	Unique Triples	Unique Entities	Relation Types	Mean Support	Median Support
Rule-Based (Patterns)	35	35	64	9	1.0	1.0
Off-the-Shelf (textacy SVO)	2,449	2,439	2,589	1,091	1.0	1.0

5.2 Precision (Manual Audit)

Method	Audited Triples	Correct Triples	Precision
Rule-Based (Patterns)	10	9	**90%**
Off-the-Shelf (textacy SVO)	10	6	**60%**

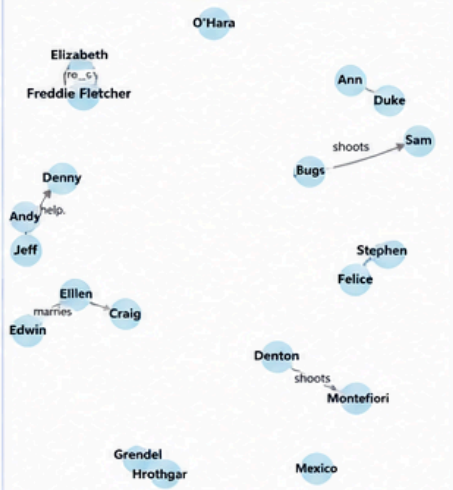
5.3 Graph Structure Metrics

Method	Nodes	Edges	Weak Components	Largest Component (# Nodes)	Density
Rule-Based (Patterns)	64	35	29	4	0.008681
Off-the-Shelf (textacy SVO)	2,589	2,439	482	1,414	0.000338

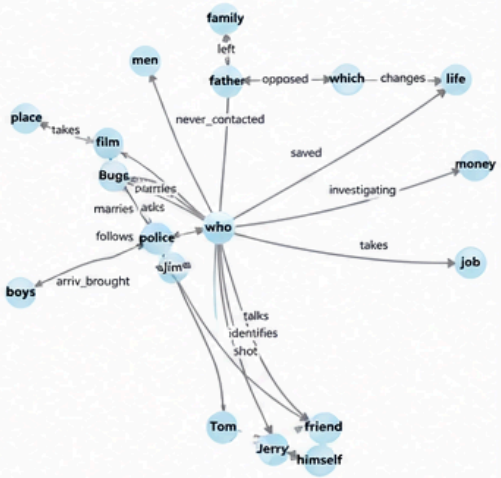
Key Insight: The off-the-shelf method extracts far more triples and relation types, but with lower precision and a very sparse graph. Rule-based extraction is much more precise but limited in coverage.

6. PRODUCED GRAPHS (Top-degree Subgraphs)

Rule-Based KG: top-degree subgraph



Off-the-Shelf textacy KG: top-degree subgraph)



Person / Character · Location · Other Entity Type

7. CONCLUSIONS

Takeaways

- Rule-based extraction achieves high precision (90%) but covers a small set of relations.
- Off-the-shelf SVO extraction dramatically increases coverage, extracting 2,439 unique triples and 1,091 relation types.
- However, lower precision (60%) and extremely low graph density indicate noisy relations.
- Generic entities (e.g., "man", "people", "police") connect unrelated plots and reduce quality.

Limitations

- No coreference resolution (pronouns ignored).
- Simple entity and relation normalization.
- Manual audit sample is small (10 triples per method).
- Narrative domain leads to sparse and fragmented graphs.

Future Work

- Improve entity filtering (remove generic terms).
- Normalize relations to reduce vocabulary sparsity.
- Use coreference resolution and better NER.
- Explore transformer-based relation extraction models.